1 Matrix Vector Multiplication

1.1 Theory

Let y describe a matrix vector multiplication. The equation would give $y = \mathbf{A}\overline{x}$, where

$$y = \begin{pmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{pmatrix} \tag{1}$$

$$y_1 = A_{11}x_1 + A_{12}x_2$$
$$y_2 = A_{21}x_1 + A_{22}x_2$$

$$y_i = \Sigma_{j=1}^{n} A_{ij}x_j \tag{2}$$

1.2 Example

Assume we are given the following and we need to apply Matrix vector multiplication

$$\begin{bmatrix} 2 & 1 \\ 4 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \tag{3}$$

$$\begin{bmatrix} 2(2) + 1(4) \\ 4(2) + 6(4) \end{bmatrix} = \begin{bmatrix} 8 \\ 32 \end{bmatrix} \tag{4}$$